and APIs for manipulating the content of responses to those requests, which is usually in HTML, XML, or JSON format. As always, we'll start with first principles and build from there. This section will provide you with an explanation of what HTML is, how to parse and modify it with PHP, and signposts where to find more information.

You might wonder why we are first learning how to manipulate existing HTML first, rather than jumping straight into building our own? The reason I chose this approach is that outputting your own web content comes with a number of implicit problems to solve beyond simply understanding how to write good markup. I want the creation of HTML to focus on things like security and good practice; those messages are important and might be diluted if you also have to wrap your head around what HTML actually is. To that end, in this section we'll learn how to take existing markup and turn it into data that can be processed in concise and efficient ways in PHP so that, when we get to the sticky business of outputting your own, you'll already have a secure foundation to build on.

Readers that are already fluent in HTML (perhaps you're coming to PHP from a front-end engineering background) and know their DOM manipulation from their complex CSS selectors can probably skim this section for the PHP specifics. Though it never hurts to review the fundamentals at any stage in your journey…

What Is HTML?

HyperText (or Hyper Text) Markup Language defines the structure (and with HTML5, the meaning) of web documents. The "Hyper" portion of the name refers to the links embedded in those documents that clients can follow to other parts of the same document or to new documents entirely. "Markup Language" means that annotations (in the form of meta-character enclosed tags) are added to the text of the document to define structure, formatting, and meaning. The tags themselves are not rendered by a client but instead instruct the client on how the document is structured and therefore how it should be rendered. The tags also impart semantic information: clients are informed about document elements such as images, tables, headers, and footers and can treat them appropriately. There are adjacent technologies to HTML to further control document appearance and behavior. Cascading Style Sheets (CSS) allows the control of things like positioning, size, fonts, and even animation of elements. JavaScript is ubiquitous as the event-driven programming language interface in web browsers to control the behavior of a document.